EMPLOYMENT PASSPORT

A Self-Sovereign, Privacy-Preserving Employment Credential Network for Bangladesh's Domestic and Overseas Labour Markets

Whitepaper - Blockchain Olympiad Bangladesh 2026

Category: Blockchain (Student) · Primary Domain: Identity & Privacy · Secondary: E-Governance · Tertiary: Fintech

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Platform: Hyperledger Fabric (permissioned) with periodic Merkle-root anchoring to a public chain

2026-08-18

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1. Executive Summary

Bangladesh sends over a million workers abroad each year and receives about $30 billion back in remittances,but the entire system runs on easily faked paperwork photocopied certificates,laminated cards, and scanned PDFs held by whoever issued them, none of which travel with the worker or can be cheaply verified by a stranger. This creates real harm: workers pay huge fees essentially to compensate for unverifiable trust, employers swap contract terms after arrival since no one witnessed the originals, experienced workers return home and get treated as beginners because their years abroad can't be proven, and honest agencies can't distinguish themselves from dishonest ones.Employment Passport solves this by putting employment facts - skill certificates, contracts, wage records, clearances - onto a shared digital system where each fact is issued as a verifiable digital credential, signed by whoever issued it (a training center, regulator, employer, or bank) and held in the worker's own digital wallet. Anyone can verify these in seconds for almost nothing, without phone calls or chasing paperwork, and the worker controls exactly what gets shared for instance, proving their income is above a certain level without revealing their employer or salary.This runs on a blockchain because the parties involved - recruiting agencies, employers, banks, governments -don't fully trust each other, and some have a direct incentive to lie (agencies want to inflate skills, employers want to deny unpaid wages). No single company or government can be trusted to hold the master record alone, so instead it's jointly maintained by the labor ministry, agency associations, banks, and international labor bodies, with the underlying technology allowing personal data to stay private while still letting the records be checked and trusted abroad.

2. The Problem

2.1 Scale and stakes

Bangladesh sent 1,130,757 workers abroad in 2025, roughly two-thirds of them to Saudi Arabia alone, and the sector delivered remittance inflows of about USD 29.6 billion in the first eleven months of that year. The composition of that outflow is the crux of the problem: official manpower data for 2025 classified 43.47% of departing workers as less-skilled and 34.46% as semi-skilled, with only 19.13% skilled and 2.94% professional. More than three-quarters of the country's most valuable export leaves without any portable, verifiable proof of what it can do.Because skill cannot be proven, it cannot be priced.

Bangladesh therefore competes on cost rather than capability. Workers absorb the difference. Official statistics put the average cost of overseas employment at roughly BDT 4.17 lakh per migrant, with male migrants paying around BDT 4.71 lakh and requiring an average of 17.6 months of foreign earnings simply to break even. World Bank and KNOMAD survey work has repeatedly identified Bangladeshi workers as paying the highest worker-borne recruitment costs in the world. Earlier IOM survey work found that the large majority of what a migrant pays never reaches a licensed agency at all. It disperses among sub-agents and intermediaries whose only product is an introduction that cannot be verified.

Meanwhile, demand-side verification is already arriving whether Bangladesh is ready or not. More than a third of Saudi-bound Bangladeshi migrants in 2025 travelled under a destination-country skill-verification programme whose certification capacity has scaled from roughly a thousand workers a month to about sixty thousand across twenty-eight approved centres. Destination states are building verification rails.

If Bangladesh **does not** build its own interoperable layer, its workers will be verified on **someone else's** terms, at **someone else's** price, with the data held by **someone else**.

2.2 Four concrete failure modes

(a) Credential fraud. Skill certificates, experience letters and trade-test results are printed documents with no machine-checkable authenticity. A forged five-year welding certificate is cheap; verifying a real one requires phoning a training centre that may not answer, in a language the employer may not speak, during working hours in a different time zone.

(b) Contract substitution and wage theft. A worker signs one contract in Dhaka and is handed a worse one on arrival. Because the original was never independently witnessed and timestamped, the worker has no admissible evidence. Unpaid or partially paid wages follow the same pattern: the only payroll record is held by the party accused of withholding.

(c) Recruitment opacity. With on the order of 2,400 licensed recruiting agencies plus an informal sub-agent layer, a prospective migrant in a rural upazila has no way to check whether an agency is licensed, what it charged previous clients, or whether the job order it is advertising exists. Fraudulent agencies are indistinguishable from honest ones at the point of sale.

(d) Non-portability and career amnesia. A returning migrant's decade of experience is invisible to domestic employers. A garment worker who moves between factories loses tenure, safety training and machine-competency records at every move. Experience that the economy paid to create is destroyed on every transition.

2.3 Stakeholders and why their incentives do not resolve this alone

Stakeholder	What they need	Why they cannot fix it unilaterally
Worker / migrant	Portable proof of skill, contract terms and wage history	Holds no institutional power; cannot compel any issuer to attest, and any self-asserted record is worthless
Recruiting agency	Cheap screening; credible differentiation from bad actors	Opacity is currently profitable for the worst actors, so the industry cannot self-regulate from inside
Destination employer	Confidence that a hire can actually do the job	Cannot audit foreign training centres; defaults to discounting all claims, which depresses wages for everyone
Government (MoEWOE / BMET)	Reduced fraud, higher-skilled export mix, protection mandate	A national database is not treated as authoritative by foreign employers and cannot bind foreign parties
Bank / MFI / insurer	Verified income for credit, insurance, remittance products	No trustworthy income signal for the informally documented; excludes millions from formal finance
Training institute (TTC)	Recognition of its certificates abroad	Individually invisible; its brand cannot travel

2.4 Why a conventional database does not solve this

The main problem with using a normal cloud database is that the records are controlled by one organization, so they can potentially be changed or deleted. No single organization, such as BMET, a private company, or an employer group, may be trusted by everyone. This is especially important when disputes happen between workers, employers, and organizations in different countries. In such cases, a record stored on one party's server may not be accepted as reliable evidence. The records also need to remain valid for many years because workers may have careers lasting decades, while companies and government agencies can close or change. Finally, workers should have ownership and control over their own records. Instead of an institution deciding who can access the information, the worker should decide what information to share and with whom. Therefore, a simple cloud database is not enough; the system needs records that are tamper-resistant, independently verifiable, long-lasting, and controlled by the worker.

3. The Solution: Employment Passport

3.1 Concept

Employment Passport gives every worker a **Decentralised Identifier (DID)** and a mobile wallet. Institutions that know something true about that worker issue **Verifiable Credentials (VCs)** into the wallet. Each credential is digitally signed by its issuer; its cryptographic hash, issuer identity, schema and revocation status are written to a Hyperledger Fabric ledger operated jointly by the network's member organisations. The credential's actual contents never touch the ledger.

A verifier - a foreign employer, a bank, a labour attaché - receives a presentation from the worker, checks the issuer's signature, checks the anchor and revocation status on-chain, and gets a yes or no in under a second. No call to Bangladesh. No trust in our servers. No exposure of data the worker did not choose to send.

3.2 Credential types

Credential	Issued by	Attests	Typical use
Identity Anchor	BMET / passport authority	Binding of DID to a government-verified expatriate identity	Root of the whole chain; issued once
Skill & Trade Credential	TTC, trade-test centre, destination-country assessor	Trade, level, assessment date, assessor	Job matching, wage banding

Credential	Issued by	Attests	Typical use
Emigration Clearance	BMET	Clearance issued, corridor, job order reference	Regulatory gate; anti-trafficking
Employment Contract	Employer + recruiting agency (dual endorsement)	Employer, role, wage, hours, term, jurisdiction	Anti-substitution evidence
Wage Payment Event	Employer payroll and/or remitting bank	Period, amount band, paid-on-time flag	Wage-theft evidence; income proof
Service Record	Employer at separation	Dates, role, conduct, reason for exit	Portable experience
Agency Conduct Record	Regulator + aggregated worker attestations	Licence status, complaint outcomes, fee compliance	Agency reputation, worker protection

3.3 Selective disclosure in practice

Credentials are issued with BBS+ signatures so that a credential holder can demonstrate an arbitrary subset of attributes in a signed credential without breaking the signature, and can prove predicates over hidden attributes. Three worked examples:

A bank that is reviewing a migrant-worker loan will pose the question: "Have you been paid at least twelve consecutive months with a wage income of at least BDT 25,000 in the past eighteen months? The worker sends a response that is a zero knowledge proof that answers yes. The bank does not find out anything about the employer, the actual salary or country.

A foreign employer asks: "is this person a certified electrician at level 3 or above, no unresolved conduct finding? The proof does not reveal the age, marital status, home district or the names of the former employers of the worker.

In a labour tribunal hearing a wage claim, the tribunal will require the worker to provide all the contract credentials. Here the worker reveals everything - and the on-chain anchor verifies that the contract terms were agreed upon prior to the worker's departure as per the on-chain agreement, which is precisely what contract substitution wipes away today.

4. Market and Partners

4.1 Market sizing

We size the market on verification events, not on population, because revenue accrues per credential issued and per verification performed.

Segment	Annual volume (order of magnitude)	Monetisable events	Notes
Outbound migration	~1.0–1.1 million departures/yr	6–9 credential and verification events per worker per cycle	Clearance, skill, contract, wage, service record, employer checks
Returning and repeat migrants	Several hundred thousand/yr	Re-verification for the next corridor; domestic re-entry	Highest willingness to pay - proven experience commands a wage premium
Domestic formal sector (RMG and manufacturing)	Workforce in the low millions with high inter-factory churn	Onboarding verification, safety-training proof, tenure	Buyer compliance audits create employer-side demand
Financial services	Millions of remittance-receiving households	Income-verification API calls for credit, insurance, NRB products	Per-call pricing; strong unit economics

Segment	Annual volume (order of magnitude)	Monetisable events	Notes
Regulator and agency tooling	~2,400 licensed agencies plus TTC network	Seat-based SaaS and per-clearance licensing	Anchor contracts; drive network effect

A conservative initial serviceable market is a single high-volume corridor. Even at a low per-event fee, one corridor's clearance and verification volume supports a self-funding operation; the network then expands corridor by corridor rather than requiring national adoption on day one.

4.2 Partner ecosystem and their incentives

Partner	Role in the network	Their incentive to join
Ministry of Expatriates' Welfare & BMET	Regulator node; issues Identity Anchor and Emigration Clearance; sets policy	Fraud reduction, higher-skill export mix, defensible worker-protection record, real-time corridor analytics
BAIRA and licensed agencies	Industry node; co-signs contracts; consumes verification	Faster placement, lower screening cost, a certified-ethical tier that commands better job orders
Technical Training Centres and assessors	Credential issuers	Their certificates become internationally checkable, raising the value of their training
Banks, MFIs, Probashi Kallyan Bank	Wage-event issuer; income-verification consumer; node operator	New creditworthy segment; lower default risk on migrant lending; remittance-product cross-sell
Destination-country regulators and employers	Verifiers; later, issuers of local skill assessments	Reduced hiring fraud and mis-skilling; alignment with existing destination-side verification programmes
ILO, IOM and civil-society bodies	Observer node; audit and standards oversight	Evidence base for fair-recruitment enforcement; independent verifiability of contract terms
Mobile operators and MFS providers	Distribution and custodial recovery rails	Reach into the migrant segment; identity-adjacent product attach

4.3 How partner incentives are allocated through the platform

Incentives are allocated as network rights and revenue shares recorded and enforced by chaincode, not as a speculative token.

Issuer revenue share. Each paid verification is split by policy: a share to the credential issuer, a share to the network operations fund, a share to the worker-protection fund. Splits are chaincode parameters changeable only by consortium vote.

Reputation as an earned asset. Agencies and employers accumulate an on-chain conduct record. High-standing members receive lower verification fees, priority in job-order matching, and an audit-lite compliance path. Standing is derived from ledger events, not from self-report, so it cannot be bought.

Regulator dividend. BMET receives comprehensive analytics and audit capabilities. By integrating this credential into protocols, regulatory bodies are empowered to monitor compliance, enforce labor standards.

Worker-side cost of zero. Workers never pay for issuance, storage or presentation. The verifier pays. This is a deliberate design constraint: any worker-borne fee reproduces the extraction the system exists to eliminate.

5. Competition and Risks

5.1 Direct competition

Competitor class	Examples	Why we differ
Blockchain career-credential networks	Velocity Network Foundation, Dock, Hyland/Learning Machine-style credentialing	Built for white-collar Western labour markets; assume employer-side HRIS integration and literate self-service. No coverage of migration clearance, contract substitution, or wage events - the failure modes that dominate this corridor.
National / destination-country verification programmes	Destination-state skill-verification and recruitment platforms; sending-state migration apps	These are single-jurisdiction and institution-custodied. We are complementary rather than competing: we integrate as an issuer and verifier, and give the worker a copy that survives across corridors.
Academic credential wallets	Diploma and micro-credential blockchain wallets, national digital-locker schemes	Cover education only. Employment, wage and conduct history - the majority of what a labour verifier needs - is out of scope.
Generic SSI toolkits	Indy/Aries-based deployments, did:web wallets	Provide primitives, not a network. The hard part is not the cryptography; it is the consortium, the regulator mandate and the corridor-specific credential schemas.

5.2 Indirect competition

Status quo paper and attestation chains - notarisation, embassy attestation, physical verification visits. Slow and expensive, but incumbent and legally familiar. This is our real competitor.

Professional background-screening firms - accurate but priced for corporate hiring, not for a construction employer screening two hundred labourers.

Social proof networks - professional networking platforms are self-asserted and irrelevant to the semi-skilled segment.

Doing nothing and discounting - employers simply assume claims are false and pay accordingly. This is rational today and is exactly the equilibrium we intend to break.

5.3 Business risks and mitigations

Risk	Severity	Mitigation
Incumbent resistance: opacity is profitable for the worst-performing intermediaries, who will lobby against adoption	High	Do not fight the middle layer - re-price it. Offer verified agencies faster clearance, a certified tier, and preferential job-order access. Anchor the mandate at the regulator and the destination-country demand side, where the rent-seekers have least leverage.
Cold start: credentials are worthless until verifiers check them, verifiers do not check until credentials exist	High	Seed the supply side with records that are already digital and already trusted - BMET clearance and TTC trade-test results - so the wallet is non-empty on day one. Launch one corridor with one large employer group rather than a national rollout.
Regulatory or political change	Medium	Consortium governance with no single controlling member; open W3C standards so credentials remain verifiable outside our network; MoUs rather than sole dependence on a single administration.
Partner non-cooperation on wage events	Medium	Dual-source design: a wage event can be issued by the employer's payroll **or** derived from the remitting bank's transaction record. No single party can veto the evidence.
Low digital literacy in the core user segment	High	USSD and agent-assisted flows; custodial recovery; branch-based enrollment. The worker's minimum required action is presenting a QR code or an ID number.

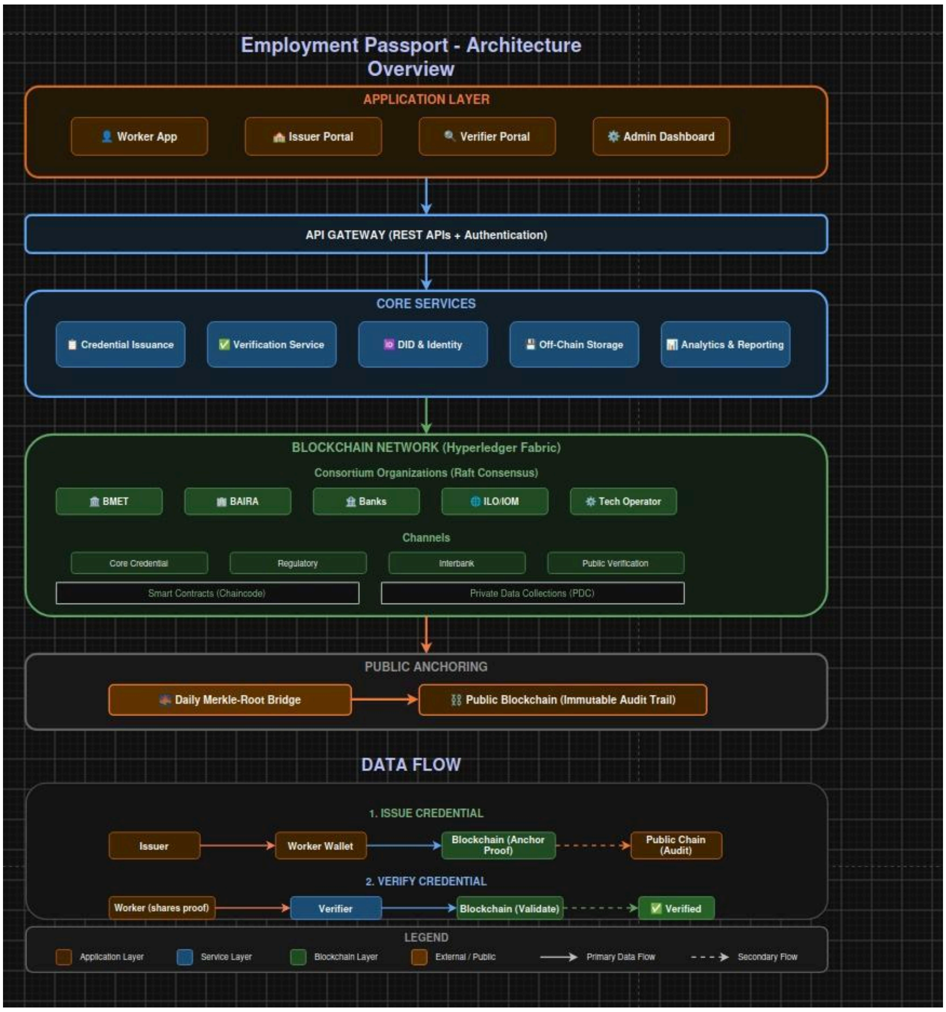
5.4 Technical risks and mitigations

Risk	Mitigation
Key loss - a worker loses their phone and with it a decade of career history	Shamir 2-of-3 social recovery split across a BMET district office, a partner bank branch and a family guardian; optional HSM-backed custodial mode for low-literacy users, with a documented path to self-custody.
Garbage in, gospel out - a colluding employer issues a false credential	Corroboration scoring. A wage claim supported by both payroll and an independent bank credit scores higher than one supported by a single issuer. Verifiers see the evidence basis, not just a boolean. Issuers stake their own on-chain reputation on every attestation.
Sybil issuers - fake employers or agencies onboarded to the network	Membership requires an MSP identity issued only after regulator-verified legal existence, with two-organisation endorsement for onboarding and an on-chain revocation path.
Privacy leakage from on-chain hashes or correlation across presentations	No PII on-chain, ever. Salted hashes only. Pairwise DIDs per verifier relationship to defeat correlation. Private data collections for anything approaching sensitive metadata.
Right-to-erasure conflict with ledger immutability	Personal data lives only off-chain under envelope encryption. Erasure is executed as crypto-shredding of the data key, leaving an unreadable ciphertext and an unlinkable hash - satisfying erasure obligations without rewriting the ledger.
Scalability under peak clearance load	Fabric with Raft ordering sustains throughput far beyond requirement (see §6.9). Verification reads are served from peer state and CDN-cached status lists, not from consensus.
Cryptographic obsolescence over a 40-year career horizon	Crypto-agility: signature suite is a schema field; credentials are re-issuable under new suites; anchors are hash-algorithm-tagged.
Node availability and operator concentration	Geographically distributed peers across independent organisations; no single organisation holds a majority of orderers; disaster-recovery peers with documented RTO/RPO in the network SLA.

6. Architecture

6.1 Architecture at a glance

The system is organised in five layers. Read top to bottom, each layer knows less about the worker than the one above it: the applications hold keys and plaintext, the services hold ciphertext and proofs, the ledger holds hashes and policy, and the public chain holds a single 32-byte number per day. That gradient is the architecture's central idea, and everything below is an elaboration of it.



Layer	Components	Responsibility	Sees personal data?
1 — Application	Worker App, Issuer Portal, Verifier Portal, Admin Dashboard	Key custody, consent, proof construction and presentation	**Yes — at the edge only** Worker device and issuer systems
2 — API gateway	REST APIs, authentication, authorisation, rate limiting	The single ingress; identity of the caller, never the content of a credential	No — ciphertext and proofs in transit
3 — Core services	Credential Issuance, Verification, DID & Identity, Off-Chain Storage, Analytics	Orchestration, schema validation, proof checking, encrypted storage, aggregate reporting	No — envelope-encrypted blobs it cannot decrypt
4 — Blockchain network	Hyperledger Fabric: 5 founding organisations, Raft ordering, 4 channels, chaincode, private data collections	Anchoring, endorsement, revocation state, schema registry, governance	**Never** Salted hashes and policy state only
5 — Public anchoring	Daily Merkle-root bridge to a public chain	External auditability of the permissioned ledger's history	Never — one 32-byte root per day

6.3 Channels and data segregation

Channel	Members	Contents
`anchor-channel`	All organisations	Credential hashes, DID document pointers, revocation status lists, schema registry
`recruitment-channel`	BMET, BAIRA, agencies, employers	Job orders, clearance events, contract endorsements
`payroll-channel`	Banks, employers, BMET	Wage-event anchors and corroboration records
`governance-channel`	Founding organisations only	Policy votes, chaincode upgrades, membership changes, fee parameters

Within channels, **private data collections** hold anything whose mere existence is sensitive - for example the corroboration detail behind a conduct finding - so that only the entitled organisations store the data while all organisations store its hash and can verify integrity.

6.4 On-chain versus off-chain

On-chain (ledger state)	Off-chain (encrypted vault)
Salted credential hash and schema identifier	Full credential JSON-LD document
Issuer DID and signature suite identifier	Worker name, NID/passport number, address, biometric references
Issuance and revocation timestamps	Contract PDF, wage slips, assessment evidence
Bitstring revocation status list	Employer HR notes and free-text conduct detail
Endorsement set and corroboration score	Any attribute subject to erasure obligations
Consent and disclosure audit hashes	Media: photographs, certificates, scanned documents

The invariant is absolute and stated as a design rule: **no personal data is ever written to the ledger, in plaintext or in reversibly encrypted form.** Off-chain storage is an IPFS-based private cluster replicated across member organisations, with per-credential envelope encryption; the data key is wrapped to the worker's DID key and to any keys the worker has explicitly granted.

6.5 Identity & Credential Layer

Workers get a digital identity (DID) anchored on the Fabric network. Verifiable Credentials (VCs) store skills, employment, and other records. Selective disclosure allows workers to share only necessary information. Credentials can be revoked without revealing which credential is being checked. Credential formats are version-controlled so their meaning cannot be secretly changed.

6.6 Key Management

Self-custody: Smartphone users keep their keys securely on their devices.

Social recovery: Keys can be recovered using 2 of 3 trusted parties (BMET, bank, family guardian).

Assisted custody: Feature-phone or low-literacy users can use a secure custodian, with worker approval and every action recorded.

Organisations use secure HSMs and key rotation to protect their identities.

6.7 Legacy System Integration

Connects with government systems for identity and migration verification. Connects with employer HR/payroll systems to record employment and salary data. Banks can provide wage-payment records through their APIs. Signed QR documents provide a paper-based fallback, allowing verification even without system integration.

6.8 Public anchoring bridge

Once daily, the Merkle root of all credential anchors committed that day is written to a public chain. The published root is a single 32-byte value containing no data and no metadata about any individual. Its purpose is narrow and important: it makes the permissioned ledger's history externally auditable, so that even a hypothetical collusion of every consortium member could not rewrite past attestations without detection by any third party holding a prior root. Public-chain fees are paid in fiat through a regulated service provider; no member of the network holds, transfers, or is exposed to a volatile digital asset.

7. Governance

7.1 Network Membership Governance

- Organisations must provide legal documents, licences, and sign the network charter to join.
- Existing members vote on admission, with the regulator having a compliance veto.
- Members can be removed for leaving, losing licences, or misconduct.
- Previously issued credentials remain valid and verifiable even if an organisation leaves.
- Roles are divided into Issuer, Verifier, Regulator, and Operator, with different permissions.
- Regular key rotation, incident response, and security audits protect the network.

7.2 Business Network Governance

- A non-profit consortium foundation manages the network rules, schemas, and common services.
- The board includes regulators, industry, banks, training organisations, and a worker representative.
- No single organisation has full control.
- Shared services include the schema registry, revocation system, wallet, SDK, and dispute-resolution support.
- Service quality and legal requirements are defined through SLAs and compliance rules.
- Disputes are handled through evidence collection, mediation, and finally the appropriate labour authority. The system provides evidence but does not decide the case.

7.3 Technology Infrastructure Governance

Each organisation operates its own network infrastructure, while government nodes remain in government facilities. Technology updates require security review, testing, and member approval. New data fields must pass a privacy review before being stored on-chain.Regular backups, audits, risk assessments, and security exercises are required. A 4-of-5 organisation agreement is needed to temporarily stop new writes during a major emergency.

7.4 Asset Tokenisation Model

The main asset is the digital employment credential, which is non-transferable and linked to the worker's DID.

It cannot be sold or transferred, reducing the possibility of credential fraud.

Each credential is unique because it represents a specific claim from a specific issuer.

There is no cryptocurrency or tradable network token; all fees are paid in normal currency.

Organisations also have a reputation score based on the reliability of their credentials. Good issuers gain reputation, while unreliable issuers lose it.

8. Valuation and Distribution

8.1 Value created

Beneficiary	Value
Worker	Lower recruitment cost through disintermediation of unverifiable middlemen; wage premium for provable skill; admissible evidence in wage and substitution disputes; access to formal credit on verified income
Employer	Screening cost and time-to-hire collapse from weeks to seconds; drastically reduced mis-hire and fraud rates
Recruiting agency	Faster placement cycles; a credible quality signal that lets compliant agencies escape competition on price alone
Government	Fraud reduction, a shift toward a higher-skilled and higher-earning export mix, live corridor analytics, and a demonstrable worker-protection record in bilateral negotiations

Beneficiary	Value
Financial sector	A new creditworthy segment with verified income; measurably lower default risk on migrant lending
National economy	Skill upgrading raises remittance per worker - the highest-leverage variable in the entire sector

8.2 Revenue model

Stream	Payer	Basis
Verification-as-a-service	Employers, agencies, verifiers	Per verification, tiered by volume; the primary stream
Platform subscription	Recruiting agencies, employers, TTCs	Annual seat-based SaaS for issuance and management tooling
Regulatory licensing	Government	Per-clearance licence fee bundled into the existing emigration-clearance process
Financial-services API	Banks, MFIs, insurers	Per income-verification call
Direct-hire marketplace	Employers	Success fee on placements made without agency intermediation
Analytics (aggregate, anonymised)	Government, development agencies, researchers	Subscription to corridor-level labour-market intelligence

8.3 Go-to-market and roadmap

Phase	Window	Objective	Success measure
0 - Prototype	Months 0–6	Fabric network with three organisations; wallet and verifier SDK; one corridor, one trade	Working end-to-end issuance and ZK verification; BCOLBD prototype round
1 - Corridor pilot	Months 6–18	Live pilot with two licensed agencies, one TTC and one destination employer group	5,000 workers enrolled; measured reduction in time-to-verify and in screening cost
2 - Regulator integration	Months 18–30	BMET clearance credential integrated into the national process; bank wage-event issuance live	100,000+ workers; clearance credential issued as standard output
3 - Destination recognition	Months 30–48	Bilateral recognition of the credential in at least one major destination corridor; domestic RMG expansion	Employer-side verification volume exceeds issuance volume - the network-effect inflection
4 - Regional	Year 4+	Interoperability with other South Asian sending states via shared W3C schemas	Cross-border credential recognition; foundation self-funding

Appendix A - Core Chaincode Interface

Function	Caller	Effect
`RegisterDID(did, docHash)`	Issuer (BMET)	Registers a DID document pointer on `anchor-channel`
`IssueCredential(credHash, schemaId, issuerDID, subjectDID, expiry)`	Authorised issuer	Anchors a credential; endorsement policy enforced per schema

Function	Caller	Effect
`RevokeCredential(credHash, reasonCode)`	Original issuer or regulator	Sets the revocation bit in the status list
`VerifyAnchor(credHash)`	Any verifier	Returns issuance state, issuer standing, endorsement set, corroboration score
`RecordDisclosure(consentHash, verifierDID)`	Worker wallet	Logs a consent event so the worker can audit who checked what
`SubmitCorroboration(credHash, sourceDID, evidenceHash)`	Secondary issuer	Raises the corroboration score of an existing claim
`UpdateAgencyStanding(agencyDID, delta, evidenceHash)`	Regulator + observer	Adjusts reputation stake; dual endorsement required
`ProposeSchema(schemaDoc)` / `VoteSchema(id, vote)`	Founding organisations	Governance-channel schema lifecycle

Appendix B - Example Credential Schema (abridged)

`EmploymentContractCredential` - fields marked ◆ are disclosable independently under BBS+; fields marked ▣ are held only off-chain.

Field	Type	Notes
`subjectDID`	DID	◆ Worker identifier; pairwise per verifier
`employerDID` / `agencyDID`	DID	◆ Both required; both must endorse
`jobTitle` / `tradeCode`	string / ISCO-08	◆ Standardised occupational code for cross-border comparability
`wageAmount` / `wageCurrency` / `wageBand`	decimal / ISO 4217 / enum	◆ Band is disclosable without the exact figure; supports ZK predicates
`contractStart` / `contractTermMonths`	date / integer	◆
`workingHoursPerWeek`, `overtimeTerms`	integer / string	◆ Key evidence in substitution disputes
`jurisdiction`, `governingLaw`	ISO 3166 / string	◆
`clearanceRef`	string	◆ Links to the BMET Emigration Clearance credential
`contractDocument`	encrypted blob ref	▣ Full signed PDF in the off-chain vault
`workerPII`	object	▣ Never on-chain in any form

Appendix D - Data Sources

Figures cited in §2 and §4 are drawn from the following public sources. Team members should re-verify each figure against the primary source before final submission, as migration statistics are revised frequently.

1. Bureau of Manpower, Employment and Training (BMET), annual overseas employment statistics, 2025 - total departures, destination distribution, and skill-category breakdown.
2. The Business Standard / Prothom Alo reporting on BMET data, 2025–2026 - year-on-year overseas employment growth, remittance inflows, and fiscal-year departure figures.
3. Bangladesh Bureau of Statistics, Survey on the Use of Remittances / cost-of-migration reporting - average migration cost and cost-recovery period.
4. World Bank / KNOMAD, Migration and Development Brief - comparative worker-paid recruitment costs.
5. International Organization for Migration, Bangladesh Household Remittance Survey - distribution of migration costs between licensed agencies and intermediaries.